



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 31.07.2026

In addition to Part I (General Handout for all courses appended to the Time table), this portion gives further specific details regarding the course.

Course No	: INSTR F311
Course Title	: Electronic Instruments and Instrumentation Technology
Instructor-in-Charge	: Sujan Yenuganti
Instructor(s)	: -----
Practical Instructors	:KM Neeraj, Kadam Aparjitha and Nimisha Bharti

1. Course Description:

An introduction to measurement techniques and instrumentation design and operation, the basic concept of units, measurement error and accuracy, Instrument design aspects, techniques and specifications of electronic instruments, the construction and design of measuring devices and measuring instruments such as Analog/ Digital meters, DSO, signal generators, signal analyzers, data loggers and recorders, electronic counters with their proper applications, Industrial Communication techniques, safety methods and instrumentation in hazardous areas. The course will also consist of laboratory practice and lab assignments.

2. Scope and Objective:

- An introduction to measurement techniques and instrumentation design and operation
- The basic concept of units, measurement error and accuracy, the construction and design of measuring devices and circuits, measuring instruments and their proper applications
- To learn about instrument design aspects, techniques and specifications of various electronic instruments
- To know various aspects of Industrial Communication techniques

3. Learning outcomes:

- Analyze the functionalities and operations of electronic measuring devices and instruments
- Know basic building blocks of electronic measuring instruments
- Excel in operating electronic measuring instruments, such as, DSO, Multimeter, Function generator, LCR meter and all other instruments.
- Know how to use electronic measuring instruments efficiently.
- Have knowledge about various industrial safety standards and bring insight into various industrial communication techniques.

4. Text books and reference books:

- T1. M.M.S.Anand, Electronic Instruments & Instrumentation Technology, PHI, 2005.
- R1. David A. bell, Electronic Instrumentation and Measurements, PHI, 2003.
- R2. Albert D. Helfrick and William D. Cooper, Modern Electronic Instrumentation and Measurement Techniques, Pearson, 2015.



5. Method of conduct of the course:

The course will be delivered by explaining theory through PowerPoint presentations and solving numerical problems on the Board. The course also includes a laboratory component alongside the theory classes.

6. Course Plan:

Module No.	Lecture Session	Reference	Learning outcomes
1:Introduction	Introduction of Electronic Instruments and Instrumentation Technology	T1	Students will be able to understand the significance of Electronic Instruments and Instrumentation Technology.
2: Analog Meters	Permanent Magnet Moving Coil Meters, DC Analog Meters AC Analog Meters, Electronic Analog Meters	R1	Students will learn the working principle of Permanent Magnet Moving Coil (PMMC) meter, DC Analog meters, AC Analog Meters.
3:Digital Meters	Digital to Analog Converter Analog to Digital Converter Digital Meters	R1	Students will learn the concept of analog to digital converters and digital to analog converters and also the working principles of digital meters
4:Analog and Digital Oscilloscope	Analog Oscilloscope Digital Oscilloscope	R1	Students will be able to understand the building blocks and working principles of analog and digital oscilloscope
5:Probes	Type of Probes Probe Loading and Measurement	T1	Students will be able to identify different types of measurement probes, and learn the effects of measurement probes on measurements
6:Bridge Instruments	Direct Current Bridges Alternating Current Bridges Application of Bridge Circuits	R1, R2	Students will be able to design signal conditioning circuits, different types of bridge instruments: DC Bridge circuits, AC bridge circuits. Students will be able to use bridge instruments for signal conditioning
7:Recorders and Data Loggers	Recorders and Data loggers	T1	Students will learn different data presentation elements, display elements, recorders, data loggers
8:Signal Generators	Sinusoidal Signal Generator Non-sinusoidal Generator Digital Signal Generator	R1	Students will learn methods of generating sinusoidal, non-sinusoidal and digital wave forms, and will be able to design circuits to generate such waves
9:Signal Analyzers	Distortion, Distortion Measuring Instruments, Spectrum Analyzer FFT Analyzer, Vector Analyzer Logic Analyzer.	R2	Students will learn the concept of distortion and noise. They will also learn different techniques and different instruments to analyze signals affected by distortion and noise.
10:Electronic Counters	Conventional Electronic Counter Reciprocal Electronic Counter, Specifications of Electronic Counter	T1	Students will learn designs and operations of different electronic counters: conventional electronic counters, reciprocal counter. Students will also be able to measure the frequency and time period of signals using counters.



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11:Grounding and Shielding	Grounding Shielding	T1	Students will understand the essence of grounding and shielding, and will be able to use grounding and shielding techniques in electronic instruments.
12:Industrial Communication Techniques	OSI network model Field bus and device network Foundation field bus	T1	Students will learn OSI layers, Network Models, Network Topologies, Interface standards, IEEE 488 (GPIB), IEEE 488.1, IEEE 488.2, HS 488, HART, Token buses and rings, Ethernet.
13:Instrumentation in Hazardous areas	Hazardous Area Classifications, Intrinsically Safe Design	T1	Students will be able to classify different hazardous areas, and will learn the approach of instrumentation in hazards areas

7. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Quiz	20 Min.	10%	-----	Closed Book
Mid-Semester exam	90 Min.	25%	<TEST_1>	Open Book
Comprehensive Examination	3 h	40%	<TEST_C>	Closed Book
Lab component	----	25%	-----	----

List of laboratory experiments:

- Study of basic components and bench instruments
- Basic op-amp circuits (I and II)
- Temperature measurement using a thermocouple, AD590 and LM35
- Temperature measurement using RTD and Thermistor
- Design of DC power supply
- Speed control of DC motor using the PWM method
- Operation of a strain gauge and its signal conditioning
- Introduction to LabVIEW software

8. Chamber Consultation Hour: To be announced during the class.

9. Notices: All notices related to the course and lab will be shared on Nalanda LMS.

10. Make-up Policy: Make-up will be given on extremely **genuine** grounds only. An application should be made in advance to request the make-up examination.

11. Note (if any):

Instructor-in-charge
Course No. INSTR F311